

MAYANK BANSAL

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SKILLS

- Data & Languages:** Python, NumPy, Pandas, Matplotlib, Seaborn, SQL, C++, Java, C
- Backend & Deployment:** Flask, FastAPI, REST APIs, Docker, Git, Render
- Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, CNNs, Neural Networks, Feature Engineering, Model Evaluation, EDA, MLOps, Deep Learning, Computer Vision, NLP, Data Science
- Vector Search & RAG:** FAISS, Sentence Transformers, LangChain, RAG, ChromaDB
- Soft Skills:** Problem Solving, Leadership, Public Speaking

INTERNSHIP

- Machine Learning Intern | Qriocity Ventures | [Offer Letter](#)**
Jun 2026 - Jul 2026
- Developed deep learning models for medical image analysis using PyTorch, TIMM, and Computer Vision techniques.
 - Constructed and trained a U-Net (ResNet34) for pneumothorax segmentation (Dice Score: 0.78) and an EfficientNet-B4 system for multi-class chest disease classification (weighted F1: 0.91).

PROJECTS

- AI Complaint Triage & Routing System | [GitHub](#)**
Jan 2026 – Feb 2026
- Automated complaint routing across 8 departments with urgency prediction using hybrid rule-based and ML confidence scoring.
 - Designed and published an 800-sample multi-class complaint dataset on Kaggle; built a modular Flask REST API with explainable decisions.
 - Implemented SQLite-based feedback storage to support prediction analysis and future model retraining.
 - Tech Stack** - Python, Flask, Scikit-learn, TF-IDF, SQLite

- Visual Search Engine | [GitHub](#)**
Feb 2026 – Mar 2026
- Built an image retrieval system using pretrained ResNet50 embeddings and FAISS nearest-neighbour search for real-time visual similarity.
 - Engineered a FastAPI backend for image preprocessing, embedding generation, and vector search with a responsive upload-and-preview frontend.
 - Optimized the end-to-end pipeline to achieve approximately **150ms retrieval latency** during inference.
 - Tech Stack** - PyTorch, FAISS, FastAPI, JavaScript

- Multi-Hop Technical Debug Assistant | [GitHub](#)**
Jun 2026 – Jul 2026
- Architected a production-style Multi-Hop RAG assistant with metadata-aware chunking, semantic retrieval, persistent FAISS indexing, and multi-document search.
 - Configured a FastAPI application with live knowledge uploads, grounded LLM responses, source attribution, retrieval scores, and performance metrics.
 - Achieved approximately **93% RAG accuracy** with an average response latency of **~800ms** using Llama 3.3 via Groq.
 - Tech Stack** - FastAPI, FAISS, Sentence Transformers, Groq (Llama 3.3)

CERTIFICATES

- Complete Data Science & ML | [GeeksForGeeks](#) Nov 2025
- AI Tools Skill Up | [GeeksForGeeks](#) Aug 2025
- LLM Engineer LaunchPad | [LPU](#) Jul 2026

ACHIEVEMENTS

- Published a technical book** — "Machine Learning Projects: A Practical Guide for B.Tech CSE Students," on Amazon Kindle and Paperback, covering 6 end-to-end ML projects across NLP, CNNs, Transfer Learning, and Audio ML | [Amazon](#)
- Published research paper** — "Mood2Mail: A Lightweight Real-Time Email Tone Analyzer Using Classical ML" on Zenodo
- Competitive programming** — 580+ problems on GeeksForGeeks (101-day streak, rank 159 / 39,000+ participants) and 160+ on LeetCode

EDUCATION

- Lovely Professional University** Punjab, India
 Bachelor of Technology - Computer Science and Engineering; **CGPA: 9.41** August 2024 - Present
- S.T. Joseph's International School** Haryana, India
 Senior Secondary (Class XII), [CBSE] | 93% May 2024